



DO-IT-YOURSELF: PROJECT

You can check the link <http://hyperphysics.phy-astr.gsu.edu/hbase/Sound/radar.html> to see how the doppler shift works between the moving targets. The speed is calculated based on the doppler shift and the relation between the frequency of waves. Fig. 6 shows the speed calculator formula for doppler radar using the frequency.

Software coding

Since we have already delved into coding in the previous ADAS part, it is assumed you already have the Raspberry Pi/Nvidia Jetson ready with an OS and Python IDLE. We had also created the code for ADAS path and the lane detection, so we carry forward the same code. Add the radar function in this very code and modify it for safe distance

marking in real time. You can refer recent EFY issues to check the process once again.

In the code, first, we import the RPI.GPIO Python module to use the GPIO and read the radar data. Then we set the pulse count and sensitivity value for the radar sensor. Next, we set the pin number on which we connect the radar. Here you can use any free GPIO on the Raspberry Pi; we have used GPIO 27. Next, create the function to count the frequency, and then implement the formula we got earlier to convert the frequency change into speed. After that, we have created the loop function using 'while' where we locate the road and detect the lane lines and also draw a safe distance marking line in real time video frame. Thereafter, we display the speed detected from the

doppler radar. Fig. 7 shows a code snippet of the ADAS and radar.

Construction and testing

Based on Fig. 2, connect the HB100 radar with LM358 and its output to the Raspberry Pi/Nvidia Jetson GPIO pin as per the circuit diagram. Connect the camera to the camera CSI port of the Raspberry Pi. Then attach the radar and camera in front of the vehicle (see Fig. 9).

After completing the connections, as shown in Fig. 8, power the device with a 5V, 2A DC adaptor and run the code you have created. It will display the speed according to the cases explained earlier and also show the safe distance. **EFY**

Ashwini Kumar Sinha, an IoT and AI enthusiast, is a tech journalist at EFY

**Wanna
Be Your
Own Boss?**
DO OPEN SOURCE.



www.opensourceforu.com

Demand for Open Source is sky rocketing. Be it for managing IT infrastructure or development of software—Open Source solutions are what customers are seeking.

All you need to do is develop expertise in an Open Source stack, and then build a team around it!

And, Open Source For You can be your friend and a guide through this journey.

TO READ OUR PRINT EDITION Visit: <https://subscribe.efyindia.com>

TO READ OUR EZONE EDITION Visit: <https://ezine.efymag.com>

WORLD'S LEADING PUBLICATION ON OPEN SOURCE

Looking for marketing solutions to engage with cutting edge techies?
Contact us at growmybiz@efy.in OR call us at +91-9811155335.

